

Roundtable

Final MVP Technology Stack, Pages & Features

Core concept: Turn participants' smartphones into a cooperative microphone network that captures group conversations, identifies speakers, generates live captions and produces a meeting summary.

Product format: Responsive web application. Participants join directly from their phone browsers without installing a native app.

1. Complete Technology Stack

A. Frontend

Component	Technology	Purpose
Framework	React 18	Build the web application
Language	TypeScript	Type-safe development
Build tool	Vite	Fast development and builds
Styling	Tailwind CSS	Responsive UI
UI components	shadcn/ui	Consistent interface
Icons	Lucide React	Interface icons
Routing	React Router	Page navigation
Audio capture	Web Audio API	Capture microphone audio
Audio recording	MediaRecorder API	Create audio chunks
Real-time client	Native WebSocket API	Send audio / receive captions
QR display	qrcode.react	Generate invitation QR codes
Hosting	Vercel	Frontend deployment

B. Backend

Component	Technology	Purpose
Language	Python 3.11	Backend and AI processing
Framework	FastAPI	REST APIs and session management
ASGI server	Uvicorn	Run FastAPI
Real-time	WebSockets	Stream audio and transcript updates
Validation	Pydantic	Validate requests/responses
Audio processing	NumPy + SciPy	Audio analysis and signal processing
Audio conversion	FFmpeg	Decode and convert audio formats
Security tokens	Python secrets	Generate secure invitation tokens
Deployment	Docker	Portable backend deployment

C. AI and Audio Processing

Task	Model / Algorithm	Purpose
Speech detection	Silero VAD	Detect speech and ignore silence
Audio quality	RMS + estimated SNR	Evaluate microphone quality

Task	Model / Algorithm	Purpose
Mic selection	Weighted quality scoring	Select clearest audio stream
Speaker attribution	Device proximity + temporal voting	Estimate who is speaking
Speech-to-text	faster-whisper	Transcribe speech
Duplicate detection	Timestamp alignment + text similarity	Avoid duplicate captions
Transcript fusion	Levenshtein + weighted consensus	Combine matching results
Speaker diarization	pyannote.audio (optional)	Handle ambiguous speaker attribution
Summary	LLM (optional)	Meeting summary and action items

D. Database and Security

Component	Technology	Purpose
Database	Supabase PostgreSQL	Sessions, participants, temporary transcripts
Authentication	Temporary session identities	No permanent accounts required
Host authorization	Secure host token	Host-only controls
Participant authorization	Temporary participant token	Session access
Invitation security	Random, expiring QR token	Time-limited joining
Transport	HTTPS + WSS	Encrypted network transport
Audio storage	Temporary server buffers	Avoid permanent raw audio storage
Cleanup	Backend expiration worker	Delete session data after retention period

E. Hosting

Frontend: Vercel — React/Vite application.

Backend: Dockerized Python server — FastAPI, WebSockets and AI models.

Database: Supabase PostgreSQL.

Version control: GitHub.

The backend is kept portable so it can be moved between servers without rebuilding the application. Free hosting should not be treated as unlimited for GPU inference, concurrent audio streams or bandwidth.

2. MVP Pages and Features

The MVP uses 8 main pages. Smaller functions such as participants and diagnostics are drawers/panels rather than separate pages.

Page 1 — Landing Page

Audience: Public

- Roundtable branding and tagline
- Create Session button
- Join Session button
- Short explanation of how Roundtable works
- Three-step usage guide
- Fixed 60-minute data deletion statement
- Responsive desktop and mobile layout

Page 2 — Create Session

Audience: Host

- Enter session name
- Set maximum number of participants
- Create session
- Generate unique session ID
- Generate secure host token
- Redirect host to session lobby
- No account registration or login

Page 3 — Session Lobby & QR Invitation

Audience: Host

- Generate shareable QR code
- Five-minute QR validity
- Live expiration countdown
- Copy/share invitation link
- Regenerate QR and invalidate old token
- Reject expired QR codes
- Close invitations when capacity is reached
- Show participants and total capacity
- Show pending join requests
- Approve or reject requests
- Remove participants
- Lock session
- Start meeting
- Enforce capacity on the backend, including simultaneous join attempts

Page 4 — Join Session

Audience: Participant

- Open session through QR or invitation link

- Display session name and available capacity
- Enter display name
- Warn about duplicate display names
- Request to join
- Wait for host approval
- Receive temporary participant ID and access token
- Show expired invitation, rejected, full and locked states

Page 5 — Microphone Setup

Audience: Participant

- Request browser microphone permission
- Select available microphone where supported
- Display live audio-level indicator
- Run a short microphone test
- Show microphone-ready status
- Select approximate seating position (optional)
- Ready button
- Keep calibration simple; no complex synchronization in MVP

Page 6 — Live Roundtable

Audience: Core feature

- Capture audio from multiple participants
- Stream audio through WebSockets
- Maintain separate device audio buffers
- Detect speech using Silero VAD
- Estimate audio quality
- Select clearest microphone for a speech segment
- Transcribe using faster-whisper
- Attribute speech using device-based scoring
- Suppress duplicate transcripts
- Display provisional captions and corrected results
- Indicate uncertain or overlapping speech
- Speaker names and timestamps
- Automatically scrolling transcript
- Live participant count
- Microphone connection status
- Leave meeting
- End meeting (host only)
- Participants drawer: view participants, connection/mic status, remove participant
- Audio diagnostics drawer: audio levels, mic quality, active streams, connection status

Page 7 — End Meeting

Audience: Host

- End meeting confirmation

- Stop all active audio streams
- Stop transcription
- Disconnect meeting WebSockets
- Mark session as ended
- Revoke invitation and participant access
- Start 60-minute retention countdown
- Redirect to meeting results
- Notify participants when host ends the session

Page 8 — Meeting Results

Audience: Host / authorized participants

- Show session name, duration, participant count, start/end times
- Transcript tab with complete speaker-attributed transcript
- Timestamps
- Search transcript
- Filter by speaker
- Copy transcript
- Download TXT
- Download PDF
- AI summary tab
- Meeting overview
- Main discussion topics
- Decisions made
- Action items
- Copy/download summary
- Show remaining retention time
- Disable server-side access after expiration
- Delete session data 60 minutes after meeting ends
- Downloaded files remain on the user's device

3. Final MVP Navigation

Landing	Create Session	Join Session
↓	↓	↓
Session Lobby / QR	Session Lobby / QR	Session Lobby / QR
↓	↓	↓
Microphone Setup	Microphone Setup	Microphone Setup
■	Live Roundtable	■
	↓	
	End Meeting	
	↓	
	Meeting Results	
	■ ■	
	Transcript Summary / Export	

4. Implementation Priority

Priority	Features
P0 — Essential	Session creation, QR joining, participant limits, microphone capture, WebSocket streaming, Whisper transcription, live captions
P1 — Core intelligence	VAD, microphone quality scoring, device-based speaker attribution, duplicate suppression
P2 — Product completion	Host controls, transcript export, summary, 60-minute server-side deletion
P3 — If time permits	Advanced diarization, improved overlapping speech handling, more sophisticated transcript fusion

Primary milestone: Get two phones in the same room to produce one correctly labelled live transcript. Once that works reliably, expand to more devices and add the remaining intelligence.